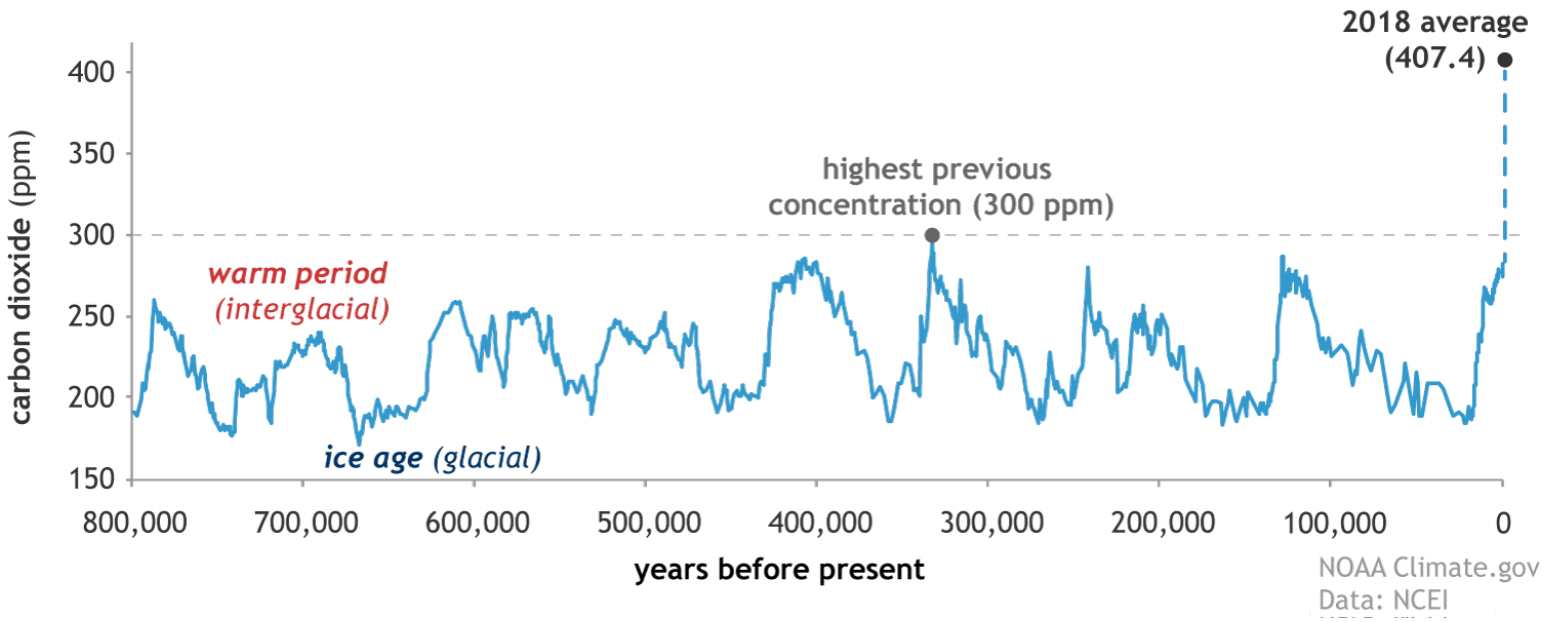
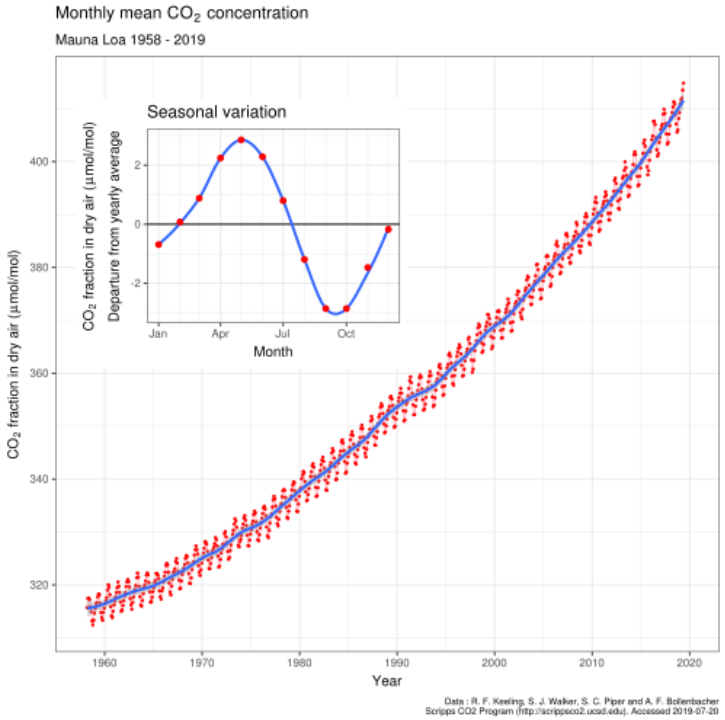
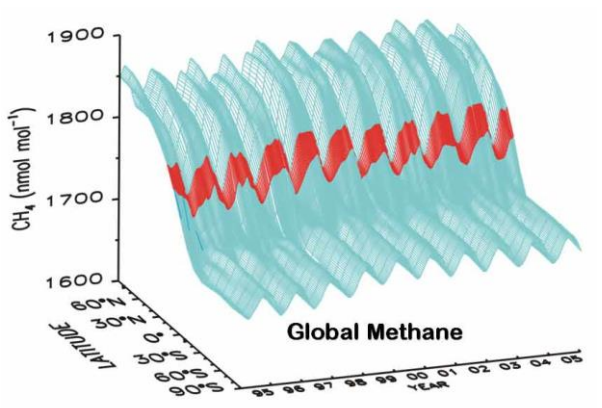
Climate Engineering or Geoengineering

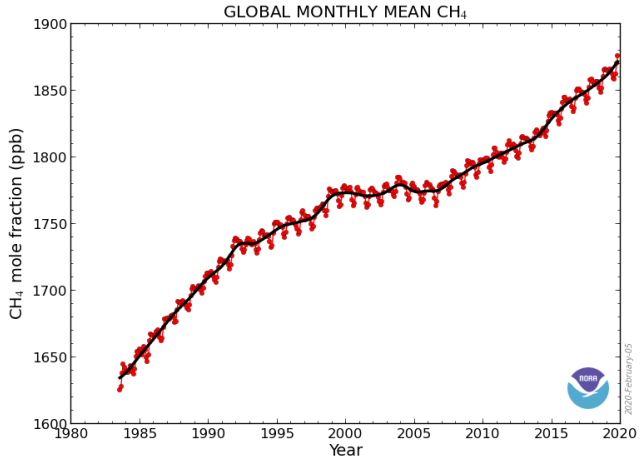
CO_2 during ice ages and warm periods for the past 800,000 years

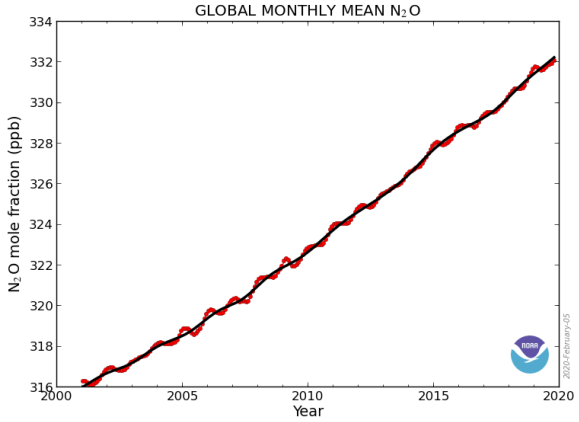


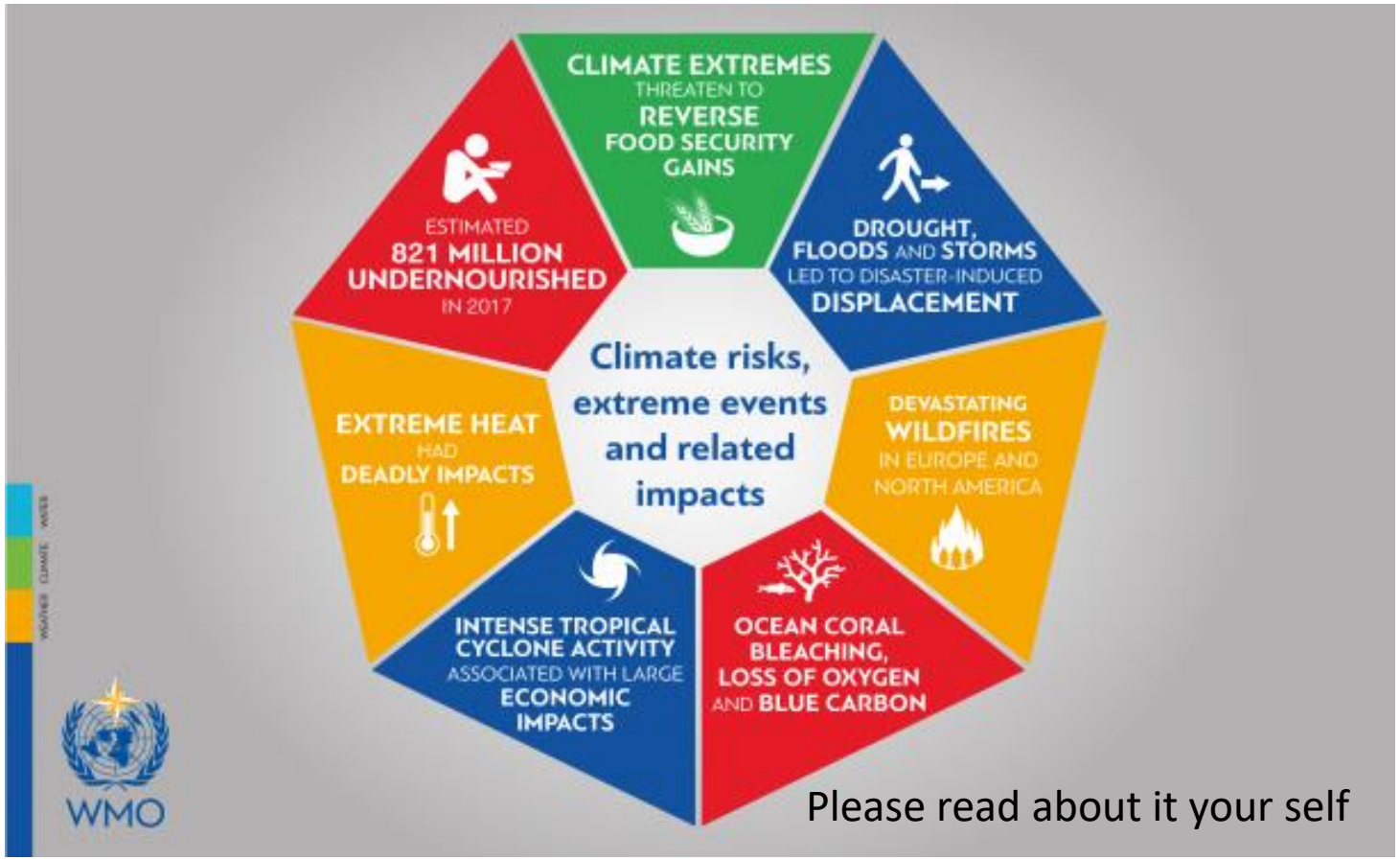


The Keeling Curve



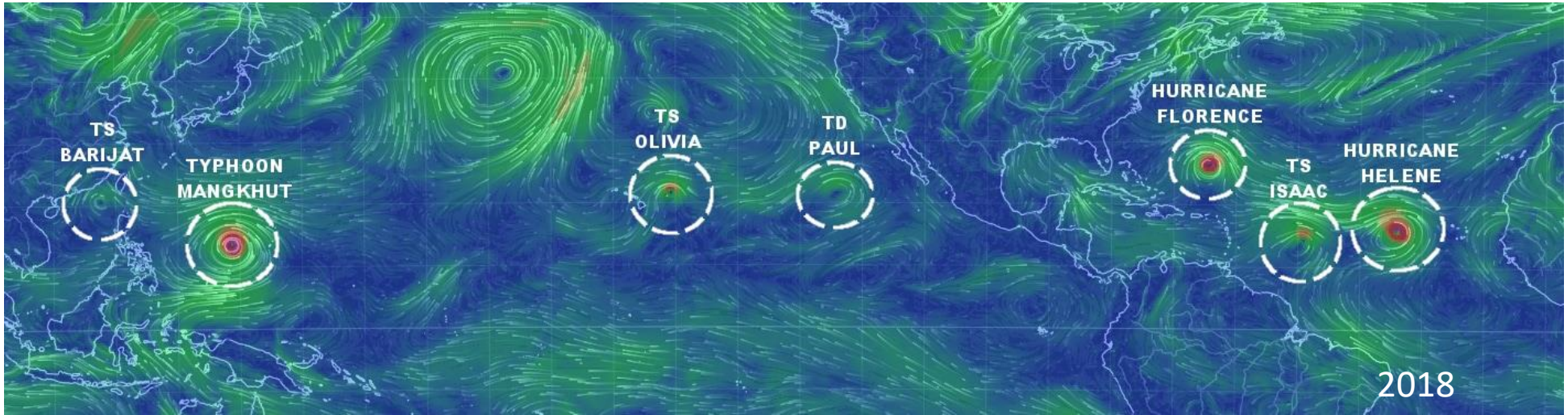






Please read about it your self

Increasing cyclones (multiple)



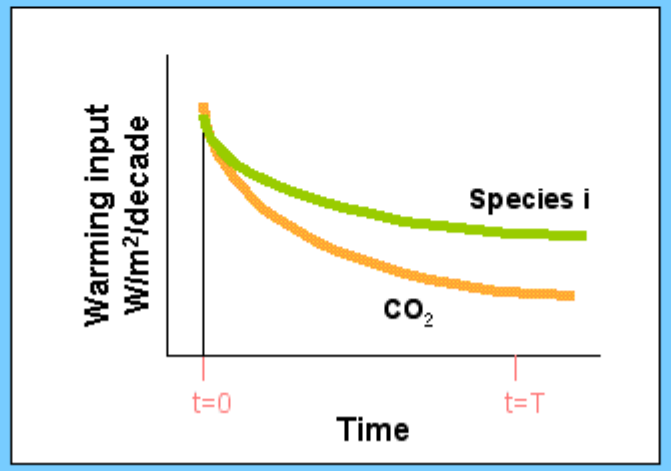
Global Warming Potential

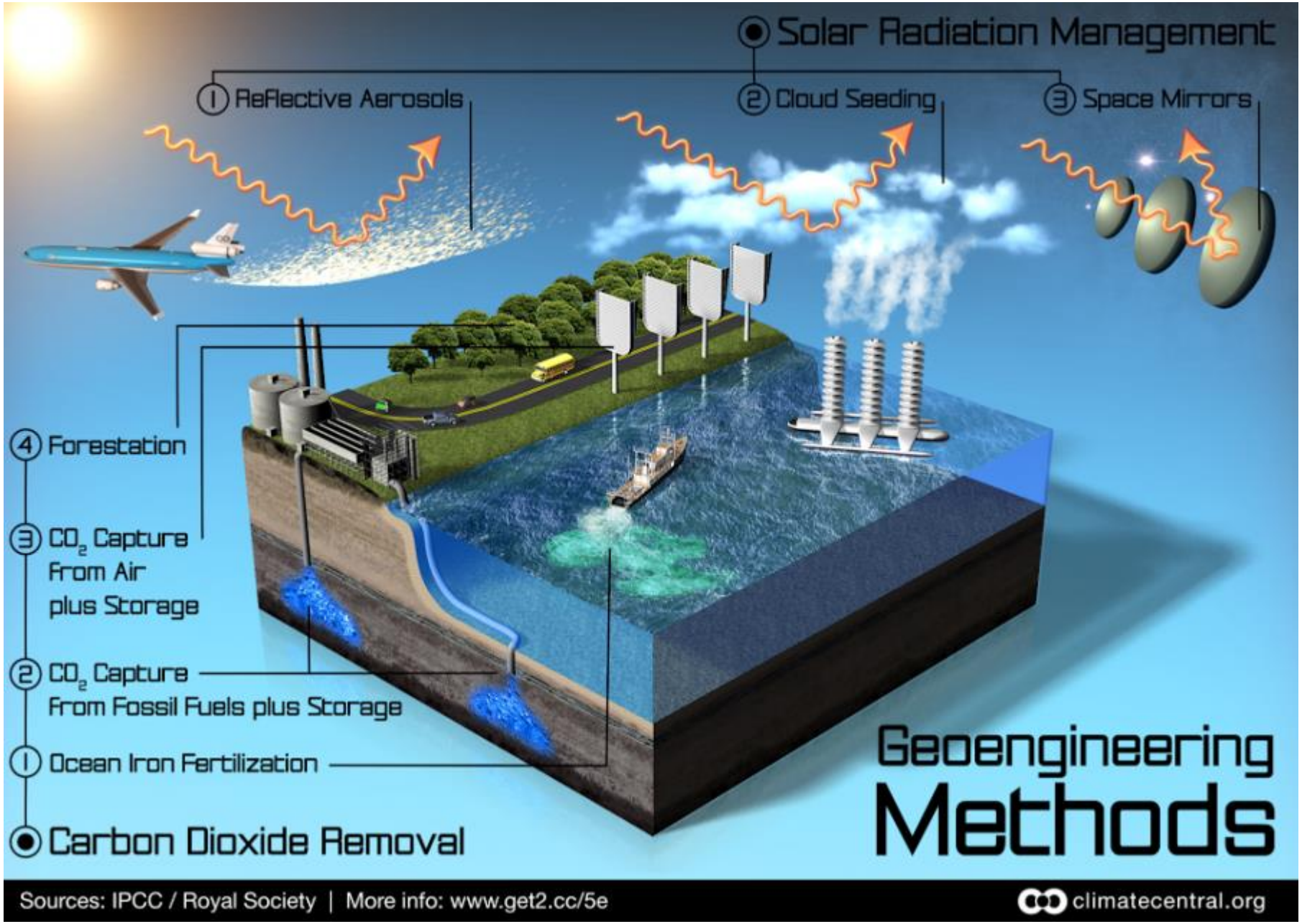
$$GWP = \frac{\int_0^n a_i c_i \, dt}{\int_0^n a_{CO_2} c_{CO_2} \, dt}$$

a_i = instantaneous radiative forcing due to unit increase in concentration of species i

Uncertainties are due to:

- Estimation of lifetime
- Radiative forcing uncertainties
- Indirect effects
- Choice of integration time, T




Solar Radiation Management
1 Reflective Aerosols
2 Cloud Seeding
3 Space Mirrors
4 Forestation
3 CO_2 Capture From Air plus Storage
2 CO_2 Capture From Fossil Fuels plus Storage
1 Ocean Iron Fertilization
Carbon Dioxide Removal
Geoengineering Methods
Sources: IPCC / Royal Society | More info: www.get2.cc/5e
climatecentral.org

Carbon dioxide removal (CDR)

CDR acts on the cause of global warming by reducing the amount of carbon dioxide in the atmosphere and storing it, thereby reducing the greenhouse effect. For example, carbon dioxide could be captured and stored by artificial trees, large-scale planting of biomass or fertilising the ocean.

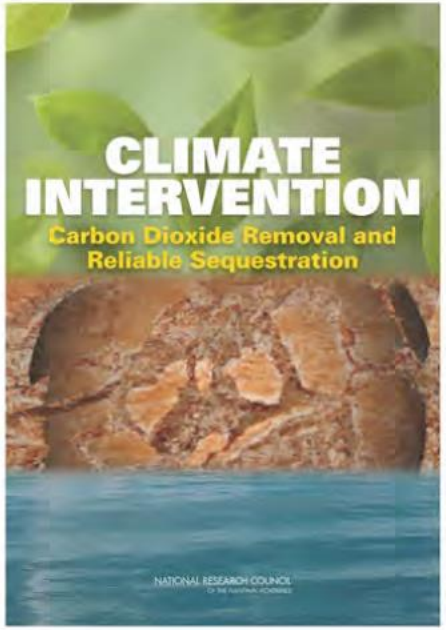
Solar radiation management (SRM)

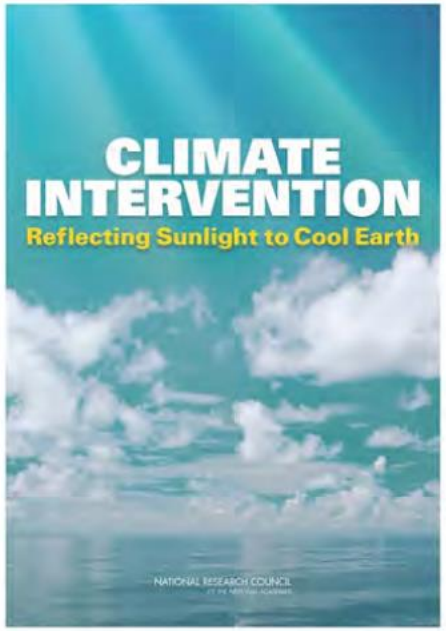
SRM aims to reflect sunlight away from the Earth, thus cooling the planet and counteracting the impact of increased greenhouse gas concentrations. Rather than acting on the cause of the problem SRM would treat one of its symptoms.
Suggested techniques have included:

- Putting massive sunshades in space.
- Injecting aerosols into the stratosphere.
- Making clouds more reflective.
- Increasing the ability of the Earth to reflect sunlight eg with reflective roofs or desert coverings.

CLIMATE INTERVENTION

Carbon Dioxide Removal (CDR)





Solar Radiation Management (SRM)

Released February 14, 2015

BOARD ON ATMOSPHERIC SCIENCES AND CLIMATE

NATIONAL RESEARCH COUNCIL
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Sponsors: U.S. National Academy of Sciences, U.S. intelligence community, National Aeronautics and Space Administration, National Oceanic and Atmospheric Administration, and U.S. Department of Energy

Stratospheric Geoengineering

Benefits

1. Reduce surface air temperatures, which could reduce or reverse negative impacts of global warming, including floods, droughts, stronger storms, sea ice melting, land-based ice sheet melting, and sea level rise
2. Increase plant productivity
3. Increase terrestrial CO_2 sink
4. Beautiful red and yellow sunsets
5. Unexpected benefits

Each of these needs to be quantified so that society can make informed decisions.

Robock, Alan, 2008: 20 reasons why geoengineering may be a bad idea. *Bull. Atomic Scientists*, **64**, No. 2, 14-18, 59, doi:10.2968/064002006.

Robock, Alan, Allison B. Marquardt, Ben Kravitz, and Georgiy Stenchikov, 2009: The benefits, risks, and costs of stratospheric geoengineering. *Geophys. Res. Lett.*, **36**, L19703, doi:10.1029/2009GL039209.

Robock, Alan, 2014: Stratospheric aerosol geoengineering. *Issues Env. Sci. Tech.* (Special issue "Geoengineering of the Climate System"), **38**, 162-185.

Risks

1. Drought in Africa and Asia
2. Perturb ecology with more diffuse radiation
3. Ozone depletion
4. Continued ocean acidification
5. Will not stop ice sheets from melting
6. Impacts on tropospheric chemistry
7. Whiter skies
8. Less solar electricity generation
9. Degrade passive solar heating
10. Rapid warming if stopped
11. Cannot stop effects quickly
12. Human error
13. Unexpected consequences
14. Commercial control
15. Military use of technology
16. Societal disruption, conflict between countries
17. Conflicts with current treaties
18. Whose hand on the thermostat?
19. Effects on airplanes flying in stratosphere
20. Effects on electrical properties of atmosphere
21. Environmental impact of implementation
22. Degrade terrestrial optical astronomy
23. Affect stargazing
24. Affect satellite remote sensing
25. More sunburn
26. Moral hazard – the prospect of it working would reduce drive for mitigation
27. Moral authority – do we have the right to do this?

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Not testable with modeling or the volcanic analog

Robock, Alan, Douglas G. MacMartin, Riley Duren, and Matthew W. Christensen, 2013: Studying geoengineering with natural and anthropogenic analogs. *Climatic Change*, **121**, 445-458, doi:10.1007/s10584-013-0777-5.

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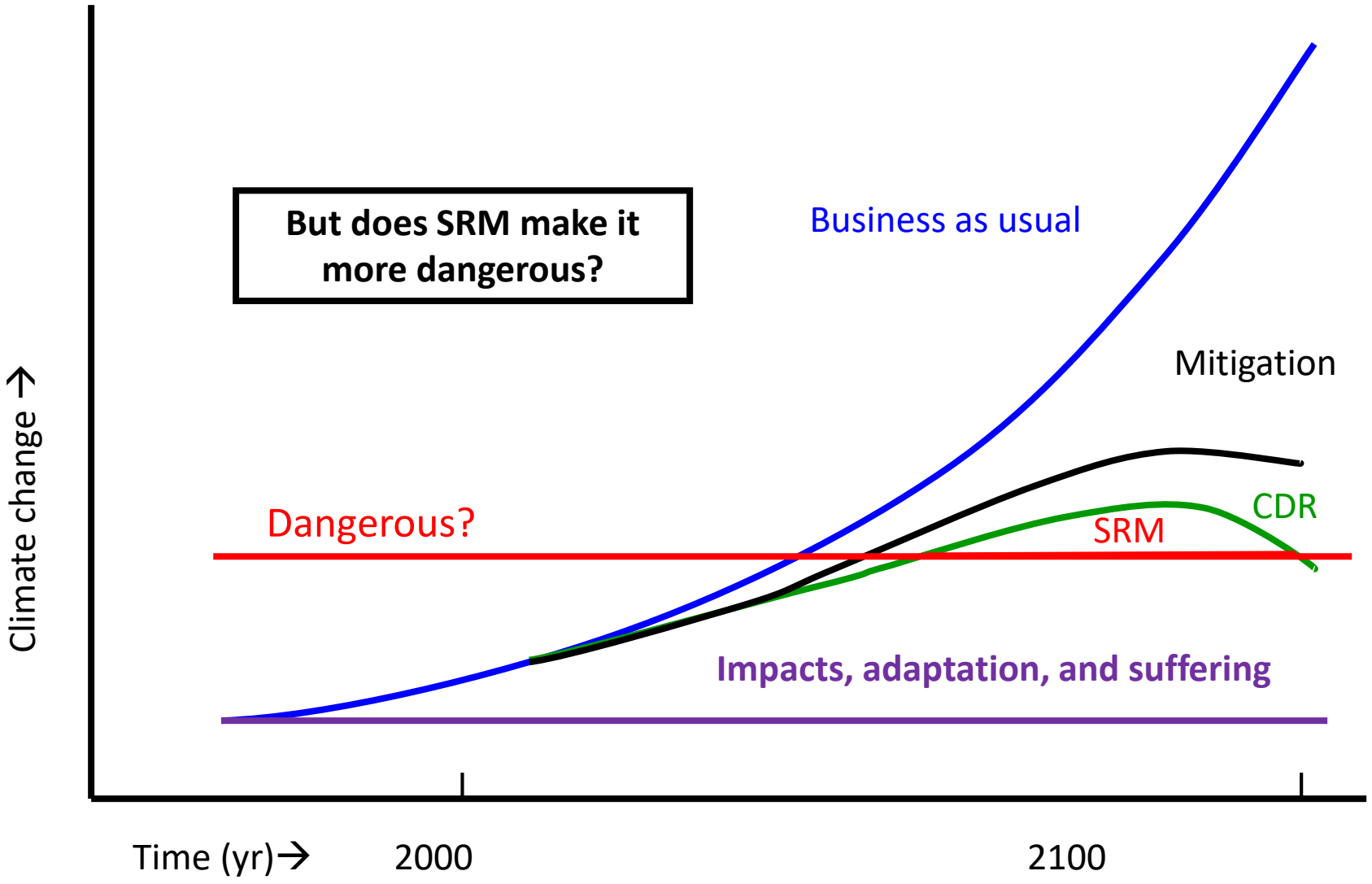


After John Shepherd's "napkin diagram"

The UN Framework Convention on Climate Change thought of “dangerous anthropogenic interference” as due to the inadvertent effects on climate from anthropogenic greenhouse gases .

We now must include geoengineering in our pledge to “prevent dangerous anthropogenic interference with the climate system.”

© New York Times, Henning Wagenbreth, Oct. 24, 2007

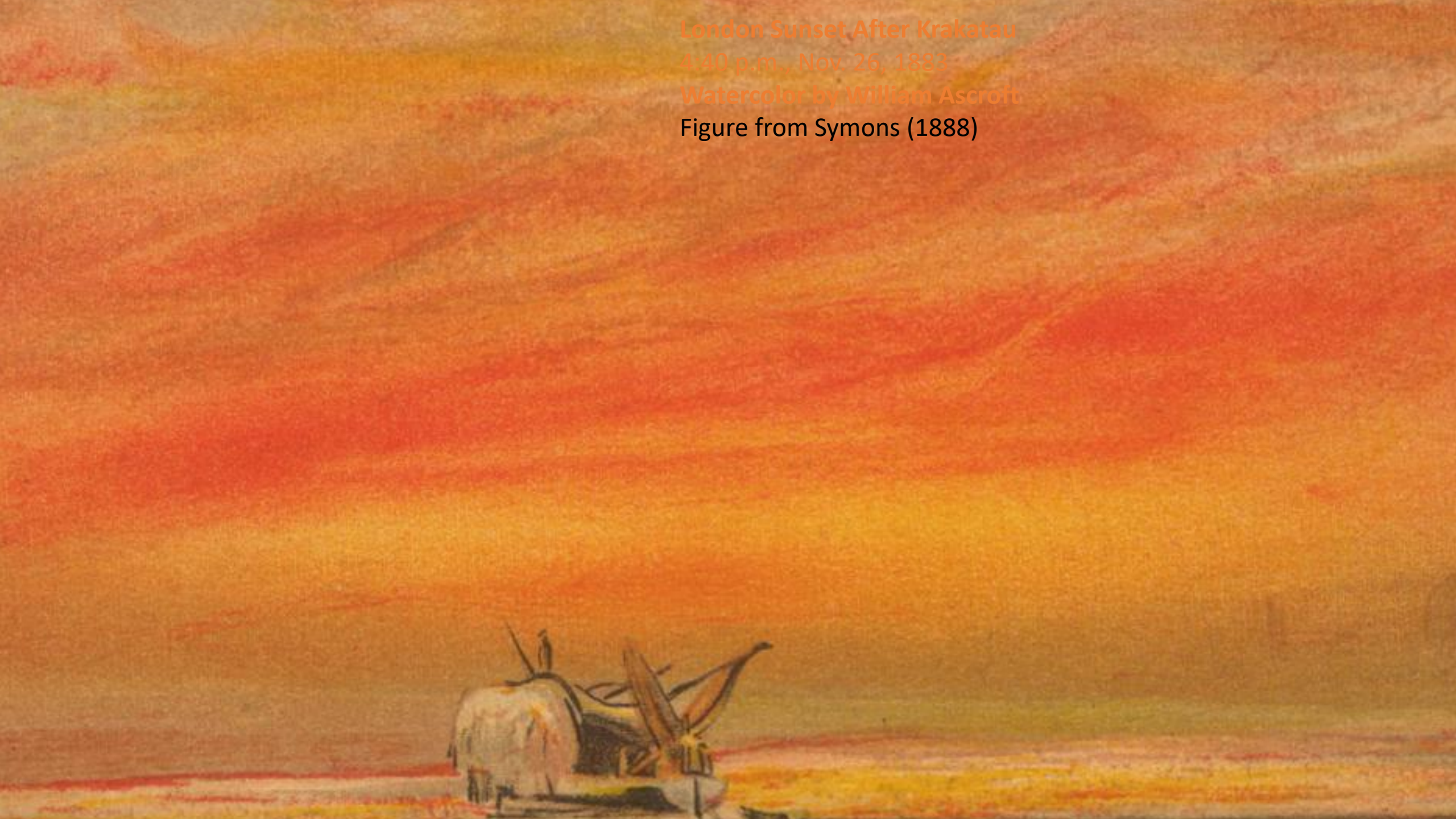
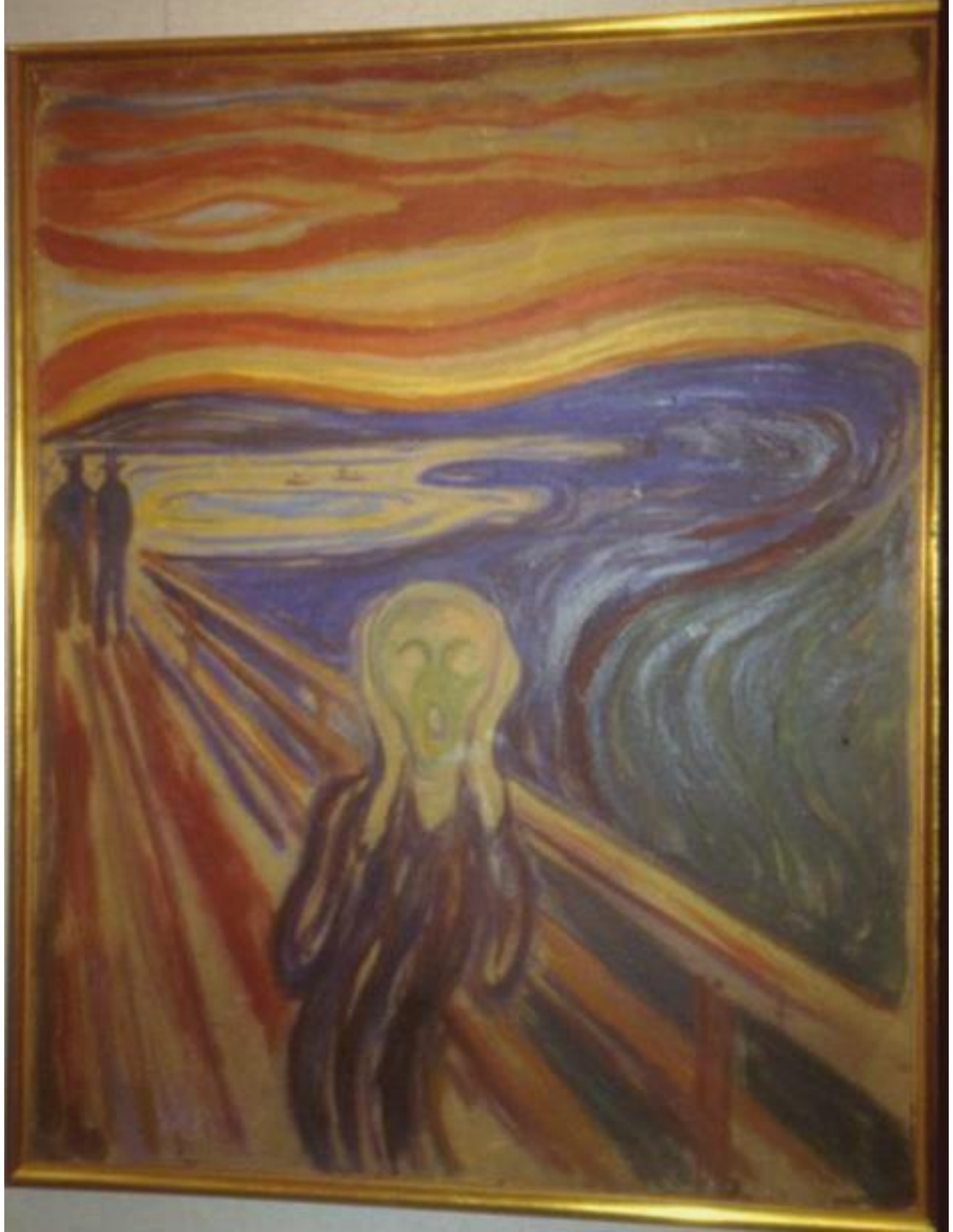


Figure from Symons (1888)

“The Scream”

Edvard Munch

Painted in 1893 based on Munch’s memory of the brilliant sunsets following the 1883 Krakatau eruption.